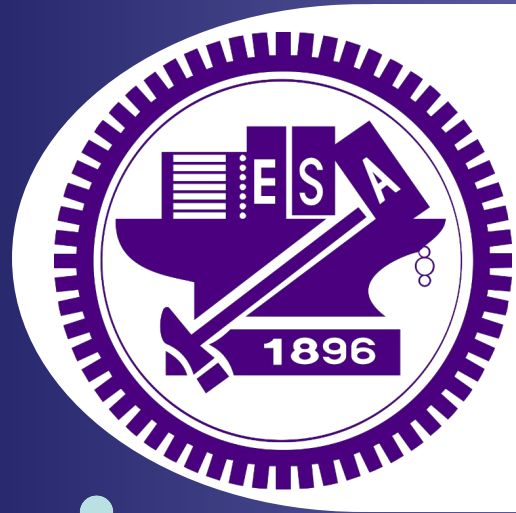




Visual Recognition using Deep Learning

2026 Spring, Final Project

Release Date: 2025/05/05 15:40

Final Project - Join a Public Kaggle Competition

- Presentation: 06/02, 06/09
- [Team List](#)
- Join the selected competition and chase the best performance
- Give a 12 mins presentation (10 mins talks + 2 mins Q&A)
- Upload the report (+codes) and slides as a team
 - 2 days before the presentation (05/31 23:59 for 06/02, 06/07 23:59 for 06/09) [late penalty: 20/day]
 - Presentation can be updated; Report **cannot** be updated.
 - Report & Code → Upload to E3 by team leader
- Grade your teammates

Selected Competitions on Kaggle (Align with presentation order)

1. Happywhale - Whale and Dolphin Identification (Classification)
2. TensorFlow - Help Protect the Great Barrier Reef (Detection)
3. NOAA Fisheries Steller Sea Lion Population Count (Detection)
4. HuBMAP - Hacking the Human Vasculature (Instance Segmentation)
5. Super Resolution in Video Games (Super Resolution)
6. SIDD Benchmark - sRGB - PSNR (Image Denoising)
7. Image Matching Challenge 2024 - Hexathlon (3D reconstruction)
8. Image Matching Challenge 2025 (3D reconstruction)

Grading Policy

- Model Performance (50%)

- Weak baseline - Get the golden medal rank: 25 pts
- Strong baseline - Achieve top-3 rank: 40 pts
- $\geq (\text{Best score} * [1.01 - 1.05])$: 50 pts

- Classification: 5% / Detection and Instance Seg: 3% / SuperRes, Denoising and Topic7: 2% / Topic8: 1%

- Presentation (30%)

- Completeness
- Innovation
- Organization
- etc.

- Reports & Code (10%)

- Within-Group Peer Review (10%)

Competition Medals				
Competition medals are awarded for top competition results. The number of medals awarded per competition varies depending on the size of the competition. Note that InClass, playground, and getting started competitions do not award medals.				
	0-99 Teams	100-249 Teams	250-999 Teams	1000+ Teams
 Bronze	Top 40%	Top 40%	Top 100	Top 10%
 Silver	Top 20%	Top 20%	Top 50	Top 5%
 Gold	Top 10%	Top 10	Top 10 + 0.2%*	Top 10 + 0.2%*

For each competition, check the score to achieve golden medal on the leaderboard!

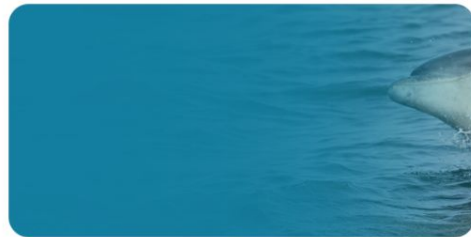
	Score	Entries	Last	Solution
	0.9132	131	4y	
	0.9043	3	4y	
	0.9028	244	4y	

Some solutions or GitHub resources may be available.
Try to reproduce and combine those solutions to outperform them.

1. Happywhale - Whale and Dolphin Identification (Classification)

Happywhale - Whale and Dolphin Identification

Identify whales and dolphins by unique characteristics



History

- # Participated Teams: 1589
- Golden metal score: 0.84909 – 0.87579 (best)

Info

- Dataset size: 62.06 GB
- Evaluation metric: Mean Average Precision @ 5 (MAP@5)
- #Teams can select this topic: 4

2. TensorFlow - Help Protect the Great Barrier Reef (Detection)

TensorFlow - Help Protect the Great Barrier Reef

Detect crown-of-thorns starfish in underwater image data



History

- # Participated Teams: 2026
- Golden metal score: 0.715 – 0.760 (best)

Info

- Dataset size: 15.23 GB
- Evaluation metric: F2 Score
- #Teams can select this topic: 4

3. [NOAA Fisheries Steller Sea Lion Population Count](#) (Detection)

NOAA Fisheries Steller Sea Lion Population Count

How many sea lions do you see?



History

- # Participated Teams: 386
- Golden metal score: 15.88009 – 10.85644 (best)

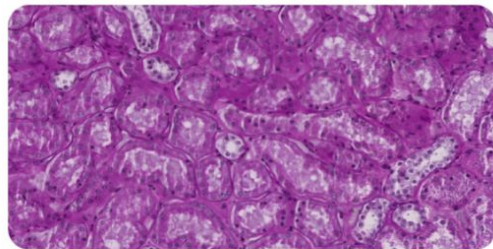
Info

- Dataset size: 103.01 GB
- Evaluation metric: RMSE
- #Teams can select this topic: 4

4. [HuBMAP - Hacking the Human Vasculature](#) (Instance Segmentation)

HuBMAP - Hacking the Human Vasculature

Segment instances of microvascular structures from healthy human kidney tissue slides.



History

- # Participated Teams: 1022
- Golden metal score: 0.544 – 0.589 (best)

Info

- Dataset size: 4.37 GB
- Evaluation metric: Average Precision over confidence scores
- #Teams can select this topic: 4

5. Super Resolution in Video Games (Super Resolution)

Super Resolution in Video Games

Try to train the SR model on a custom gaming dataset collected using Unreal Engine



History

- # Participated Teams: 38
- Golden metal score: 33.132 – 33.80 (Best)

Info

- Dataset size: 51.4 GB
- Evaluation metric: PSNR
- #Teams can select this topic: 4

6. SIDD Benchmark - sRGB - PSNR (Low-Level Vision)

SIDD Benchmark - sRGB - PSNR

Image denoising benchmark on real sRGB images using PSNR metric



History

- # Participated Teams: 98
- Golden metal score: 40.0019 – 40.7423 (best)

Info

- Dataset size: SIDD Medium (~12 GB)
- Evaluation metric: PSNR
- #Teams can select this topic: 4

7. Image Matching Challenge 2024 - Hexathlon (3D reconstruction)

Image Matching Challenge 2024 - Hexathlon

Reconstruct 3D scenes from 2D images over six different domains



History

- # Participated Teams: 930
- Golden metal score: 0.178508 – 0.259368 (best)

Info

- Dataset size: 3.97 GB
- Evaluation metric: mean Average Accuracy (mAA)
- #Teams can select this topic: 4

8. Image Matching Challenge 2025 (3D reconstruction)

Image Matching Challenge 2025

Reconstruct 3D scenes from messy image collections.



History

- # Participated Teams: 943
- Golden metal score: 43.34 – 56.00 (best)

Info

- Dataset size: 5.22 GB
- Evaluation metric: mean Average Accuracy (mAA)
- #Teams can select this topic: 4

Important Dates

Event	Date	Note
Select Topic/Competition	05/05 - 05/11	Fill out the competition selection form by team leader . Topics will be assigned based on the order of form submission and each team's topic priorities . After you submit the form, we'll email your team's assigned topic to the team leader within 2 days.
Final Presentation I	06/02	1. Happywhale - Whale and Dolphin Identification (Classification) 2. TensorFlow - Help Protect the Great Barrier Reef (Detection) 3. NOAA Fisheries Steller Sea Lion Population Count (Detection) 4. HuBMAP - Hacking the Human Vasculature (Instance Segmentation)
Final Presentation II	06/09	5. Super Resolution in Video Games (Super Resolution) 6. SIDD Benchmark - sRGB - PSNR (Image Denoising) 7. Image Matching Challenge 2024 - Hexathlon (3D reconstruction) 8. Image Matching Challenge 2025 (3D reconstruction)

- Link to the [competition selection form](#) (login required)
- You **CANNOT** change topic after submitting / being assigned, pick up topic early and carefully.
- After 05/12, we will assign a topic for you if your team haven't filled the form.
- **NOTE: Some teams for topic4 (HuBMAP) may be assigned to make presentation on 0609.**
- **NOTE: Some teams for topic5 (Super Resolution) may be assigned to make presentation on 0602.**

About the report and presentation

Presentation should contain the following sections / contents

- Introduction
- Related works
- Method / Approach
- Experimental Results
- Conclusion
- Reference

162...	Gaurav Gooner Roy		0.56459	1	22d
162...	jimmy15923		0.56459	1	now

Your First Entry ⬆

Welcome to the leaderboard!

Your score represents your submission's accuracy. For example, a score of 0.7 in this competition indicates you predicted Titanic survival correctly for 70% of people.

What next? You've got a few options:

- 🧠 Learn skills that can improve your score in our [Intro to Machine Learning course](#) by Dan Becker.
- 🗣 Check out the discussion forum to find lots of tutorials and insights from other competitors.
- 🏆 Find a new challenge by entering one of our [open, active competitions](#) or searching our [public datasets](#).

Also, don't forget to screen shot your rank in the report/presentation

Final Project Remarks

- Check the requirements: [Presentation Tips](#)
- Upload your presentation slide to [Google Drive Here](#)
 - The upload filename should be **Group[GROUP_ID]_[TOPIC]_[LEADER_STUDENT_ID]**
 - **GROUP_ID must be a two-digit number**, padded with a leading zero if needed.
 - e.g. Group00_Happywhale - Whale and Dolphin Identification_123456789.pptx
 - e.g. Group99_NOAA Fisheries Steller Sea Lion Population Count_112233445.pptx
- REMIND AGAIN: Must include additional info in your presentation
 1. **Kaggle performance snapshot (see p14)**
 2. **Team member contribution table (see p16)**
 3. **Link to your code**

	[StudentID 1]	[StudentID 2]	[StudentID 3]	[StudentID 4]
Literature survey	[0 - 100] %			
Approach design				
Approach implementation (experiment)				
Report writing				
Slide making and oral presentation				

An example

	0856065	0856078	0856005
Literature survey	100	0	0
Approach design	0	50	50
Approach implementation (experiment)	0	30	70
Report writing	80	20	0
Slide making and oral presentation	33	33	33

Good Luck!

